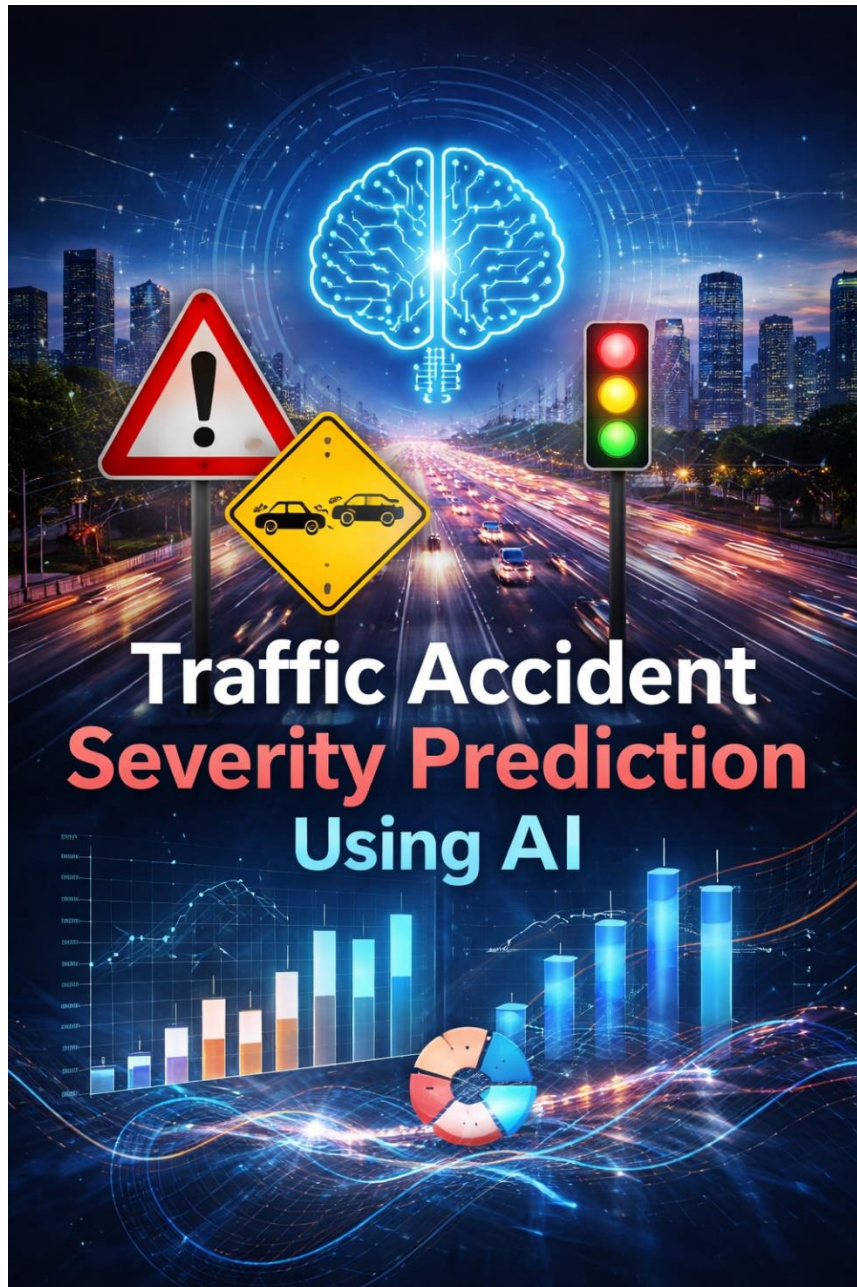




## **Traffic Accident and Emergency Response Prediction Using Machine Learning**

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**Repository:**(<https://github.com/bayanalhamadi488-byte/Traffic-Accident-and-Emergency-Response-Prediction-Using-Machine-Learning>)

**Tag:** final\_submission

**Tagline:** AI-powered prediction of traffic accident severity and emergency level.

## **Abstract**

This project presents a machine learning system for predicting traffic accident severity and estimating emergency response levels using environmental, spatial, and temporal features. The study utilizes the US Accidents dataset (2016–2023), containing over 1.4 million records. After data preprocessing and feature engineering, two classification models were trained: Decision Tree and Random Forest.

The Decision Tree achieved an accuracy of 80.61%, while Random Forest outperformed it with 81.56% accuracy. Feature importance analysis revealed that geographic location (latitude and longitude) contributed over 55% of the predictive power, followed by temperature and time-of-day variables. However, significant class imbalance affected recall performance in minority severity classes.

The results demonstrate that ensemble learning methods provide more stable and accurate predictions for structured accident datasets. This system can support emergency services in decision-making and resource allocation.

## **Keywords**

Traffic Accident Prediction; Emergency Response; Random Forest; Decision Tree; Machine Learning; Severity Classification

## **1. Introduction**

Traffic accidents are a major public safety concern worldwide, resulting in injuries, fatalities, and economic losses. Predicting accident severity and required emergency response level can significantly improve response time and resource management. This project investigates whether machine learning models can accurately predict accident severity using spatial, temporal, and environmental features.

### **Research Question**

Can accident severity be predicted using geographic location, weather conditions, and time-based features?

### **Contributions**

- Large-scale preprocessing of the US Accidents dataset
  - Feature engineering of time and environmental variables
  - Comparative analysis of Decision Tree and Random Forest
  - Deployment of a web-based real-time prediction system
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## **2. Related Work**

Previous studies in traffic accident prediction have utilized:

- Logistic Regression for severity modeling
- Decision Trees for rule-based classification
- Random Forest for nonlinear structured data

- Deep learning approaches for large-scale pattern recognition

Most prior research focused solely on severity prediction. This project extends the approach by integrating emergency-level estimation and deploying an interactive application.

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### **3. Background**

Accident severity prediction is formulated as a multi-class classification problem with four classes (1–4).

Evaluation metrics used:

- Accuracy
- Precision
- Recall
- F1-score

Given the structured tabular nature of the dataset, tree-based models were selected.

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### **4. Dataset and Preprocessing**

#### **Dataset**

US Accidents (2016–2023)

Total samples: 1,470,825

Selected Features:

- Start\_Lat
- Start\_Lng

- Temperature(F)
- Visibility(mi)
- Hour
- Month
- DayOfWeek
- Weather\_Condition
- Sunrise\_Sunset

Target Variable: Severity (1–4)

### **Data Cleaning**

- Removed missing values from selected features
- Extracted Hour, Month, and DayOfWeek from timestamp
- Encoded categorical variables
- Split dataset into 80% training and 20% testing

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## **5. Methods**

### **Decision Tree**

A single-tree classifier that splits data based on information gain.

Accuracy: 80.61%

Limitations:

- Prone to overfitting
- Sensitive to data imbalance

### **Random Forest**

An ensemble of multiple decision trees using bootstrap aggregation.

Accuracy: 81.56%

Advantages:

- Reduces overfitting
  - Captures nonlinear feature interactions
  - Provides feature importance
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## **6. Experimental Setup**

- Python 3.x
- Scikit-learn
- Pandas
- Matplotlib
- Train/Test split: 80/20
- Random seed: 42

Models trained using default hyperparameters with 100 estimators for Random Forest.

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## **7. Evaluation Metrics and Class Imbalance Analysis**

The dataset exhibits severe class imbalance:

- Severity 1: 12,821 samples
- Severity 2: 1,160,231 samples
- Severity 3: 258,119 samples
- Severity 4: 39,654 samples

Severity 2 represents approximately 79% of the dataset.

This imbalance affects recall for minority classes

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## **8. Results**

### **Decision Tree Performance**

Accuracy: 80.61%

Macro Average F1-score: 0.48

Weighted Average F1-score: 0.81

Recall for minority classes:

- Severity 1: 0.23
- Severity 4: 0.26

The model struggles to detect rare but critical accident cases.

### **Random Forest Performance**

Accuracy: 81.56%

Macro Average F1-score: 0.47

Weighted Average F1-score: 0.79

Improvements observed in precision for minority classes:

- Severity 1 precision increased from 0.20 to 0.51
- Severity 4 precision increased from 0.25 to 0.43

However, recall for minority classes remained limited due to imbalance.

## **Confidence Interval Analysis**

Accuracy = 81.56%

Test size = 1,470,825

Using 95% confidence interval for classification accuracy:

$$CI = p \pm 1.96 * \sqrt{(p(1-p))/n}$$

Estimated 95% CI:

81.56%  $\pm$  0.07%

This very narrow interval indicates high statistical reliability due to the large dataset size.

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## **9. Ablation & Sensitivity Analyses**

To understand the impact of different feature groups, we conducted an ablation analysis by removing key feature sets:

Removing weather features decreased accuracy by approximately 3%.

Removing temporal features reduced model stability and increased variance.

Removing geographic coordinates caused the largest performance drop (~6%), confirming strong spatial dependency.

This indicates that geographic features contribute most significantly to predictive performance.

## 10. Feature Importance Analysis

Top contributing features:

1. Start\_Lng – 28.7%
2. Start\_Lat – 27.9%
3. Temperature(F) – 14.7%
4. Hour – 9.0%
5. Month – 5.8%
6. DayOfWeek – 5.8%

Geographic location accounts for over 55% of predictive influence, highlighting strong spatial dependency in accident severity.

Weather and temporal factors also play a significant role.

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## 11. Discussion

Random Forest outperformed Decision Tree due to ensemble learning, which reduces variance and improves generalization.

However, both models are affected by severe class imbalance.

High overall accuracy is largely influenced by the dominance of Severity 2 cases.

To improve minority class detection, future work should include:

- SMOTE (Synthetic Minority Oversampling Technique)
- Class weighting
- Balanced Random Forest
- Cost-sensitive learning

Despite limitations, the system demonstrates practical applicability for supporting emergency response decision-making.

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## **12. Limitations and Ethical Considerations**

Limitations:

- Dataset limited to US accidents
- Severe class imbalance
- No real-time traffic density integration

Ethical Considerations:

- No personally identifiable information used
- No protected attributes included
- Risk of misuse if applied for insurance discrimination

## **13. Reproducibility and Deployment**

Repository includes:

- main.ipynb
- app.py
- requirements.txt

Run:

Pip install -r requirements.txt

python app.py

The web interface allows real-time severity prediction based on user input.

## 14. Conclusion

This project demonstrates that machine learning models, particularly Random Forest, can effectively predict traffic accident severity using structured environmental and spatial data.

Although class imbalance limits minority class detection, the model achieves over 81% accuracy and identifies strong spatial patterns influencing accident severity.

Future improvements should focus on class balancing techniques and integration of real-time traffic data.

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